

Medical Diagnosis Expert System

Decision Tree Reference

31 conditions across 7 symptom branches

How to read these trees

The system is a **forward-chaining rule base**, not a literal decision tree. The question-ordering strategy, however, induces a tree structure over the knowledge base, and drawing it that way makes the discriminative logic far easier to audit.

Five screening items — the **fever type**, plus **red eyes**, **shortness of breath**, **fatigue**, and **loss of appetite** — decide which branch or branches open. Each branch then asks only the symptoms belonging to its candidate conditions and scores the affirmative answers against a threshold.

Branching is shared, not linear. Where two conditions begin with the same symptom, they share a node and separate only at the level where their presentations actually diverge. In the fever branch, *fatigue* roots three conditions and *chest pain* roots two. That shared structure is the point of the diagram.

Paths are representative, not exhaustive. A condition is reported when *enough* of its symptoms are present, not only when one exact path is walked end to end. The tree shows the discriminative structure; the scoring threshold governs the actual firing.

Node colours

- Symptom question
- Condition
- Screening question

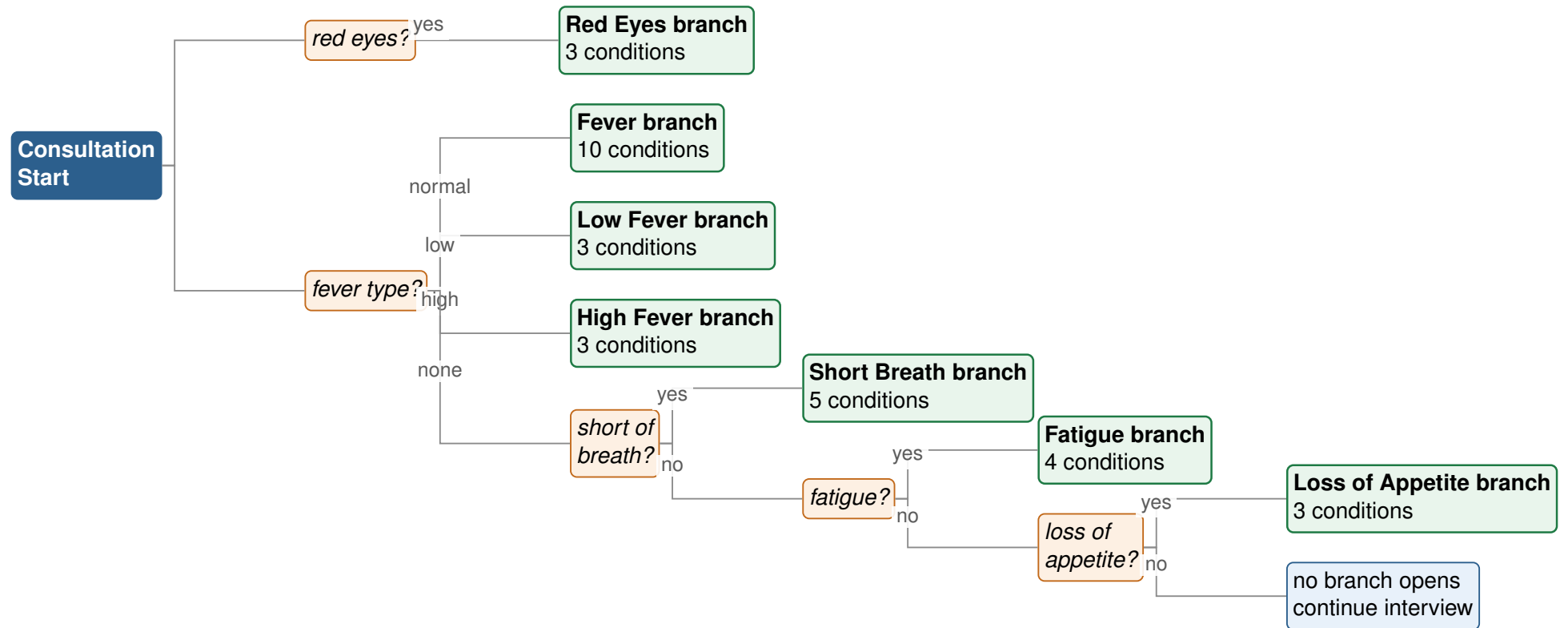
Branch summary

Branch	Conditions	Page
Fever	10	3
Low Fever	3	4
High Fever	3	5
Short Breath	5	6
Fatigue	4	7
Loss of Appetite	3	8
Red Eyes	3	9
Total	31	

Branch selection is shown on page 2. The red-eye branch is evaluated independently of the fever path, so both may open in the same consultation.

Branch Selection

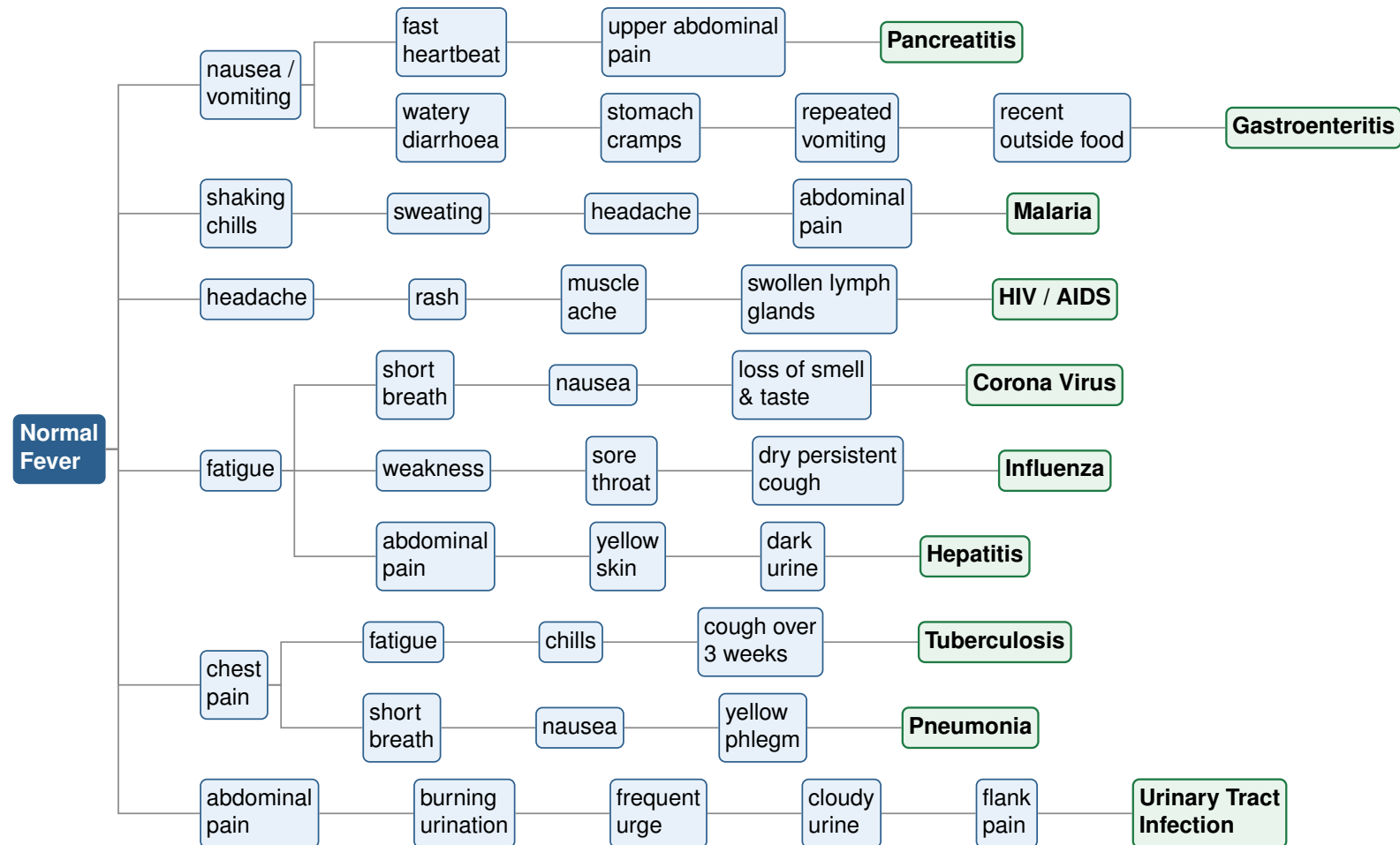
Five screening items | Routes to 7 branches



The fever path is evaluated by precedence: fever type first, then shortness of breath, then fatigue, then loss of appetite. If none of these open a branch, the interview continues rather than concluding.

Fever

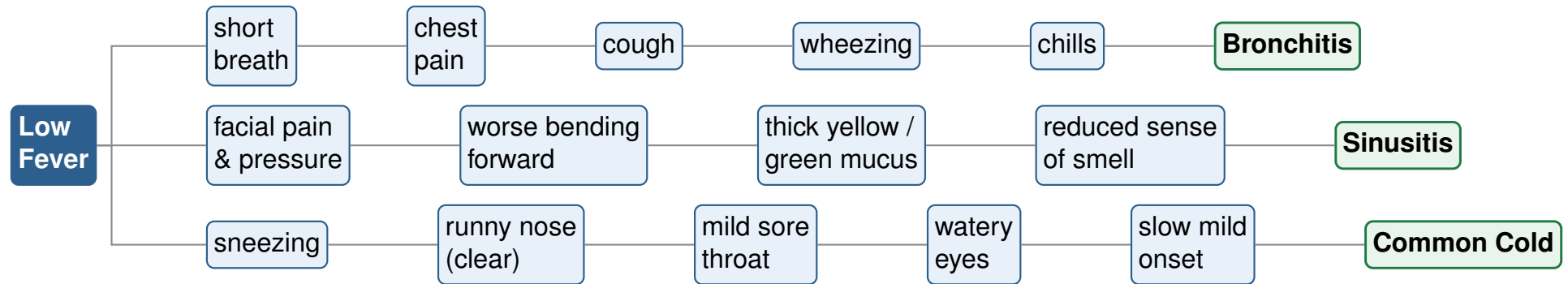
Normal fever branch | 10 conditions



The largest subtree. Three symptoms root families of conditions rather than single paths: **nausea/vomiting** (pancreatitis, gastroenteritis), **fatigue** (corona virus, influenza, hepatitis), and **chest pain** (tuberculosis, pneumonia). Each family separates one level down.

Low Fever

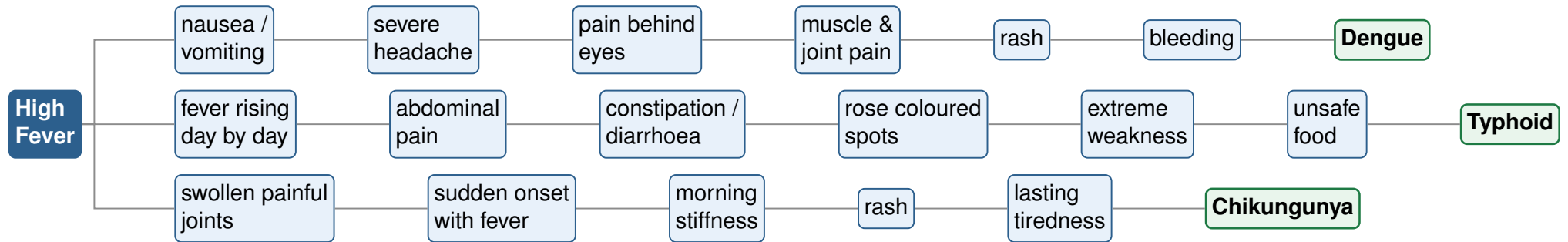
Low fever branch | 3 conditions



Three disjoint paths with no shared prefix — the first symptom alone discriminates fully here. Fever grade is doing real work: the same reported fever routes to an entirely different condition set at low grade than at normal or high.

High Fever

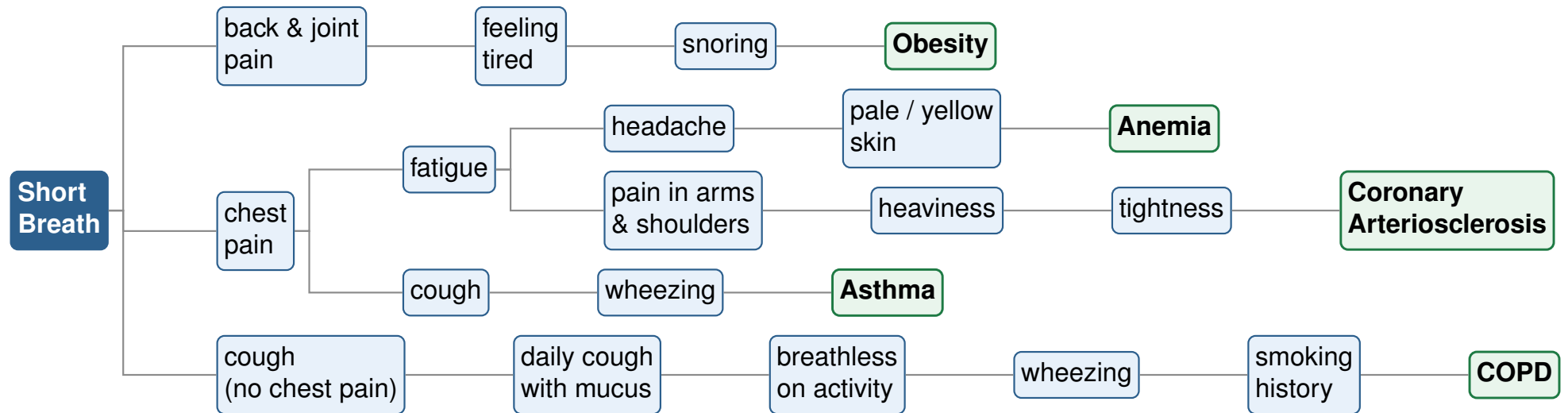
High fever branch | 3 conditions



The deepest paths in the knowledge base, at six levels. Dengue and chikungunya share both fever and rash, so discrimination rests on the later nodes — pain behind the eyes and bleeding for dengue, joint swelling with morning stiffness for chikungunya.

Short Breath

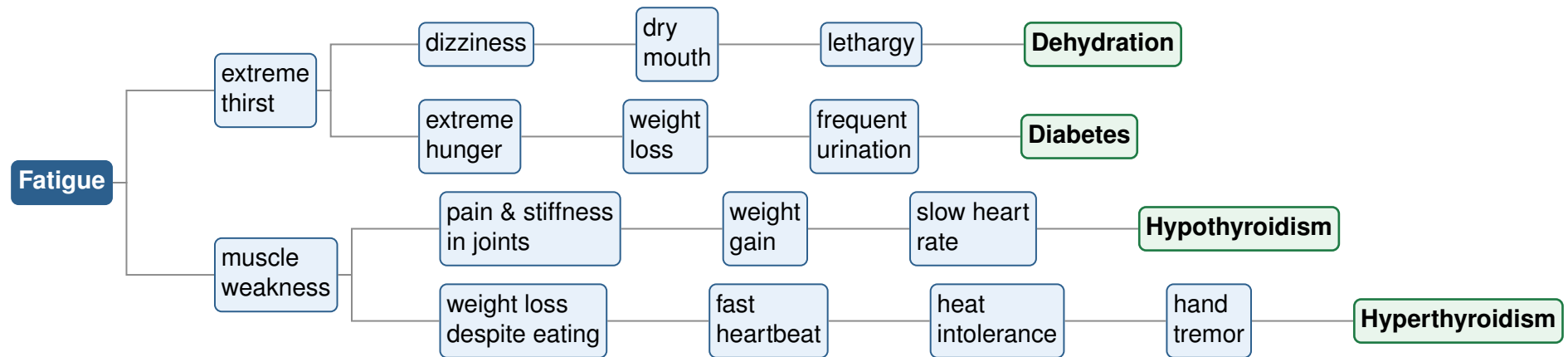
No fever | 5 conditions



The only branch with a three-level split. **Chest pain** separates the airway path (asthma) from the systemic path, and **fatigue** then separates anemia from coronary arteriosclerosis. Asthma and COPD are deliberately separated at the root by whether chest pain is present, since both otherwise present with cough and wheezing.

Fatigue

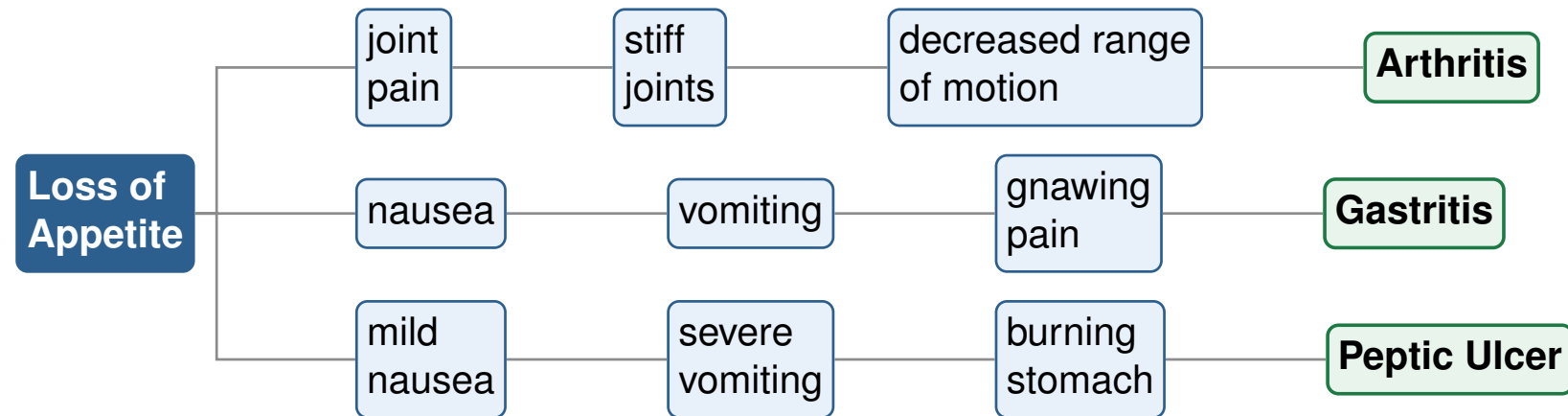
No fever, no short breath | 4 conditions



The cleanest binary structure in the knowledge base: two roots, each splitting in two. **Extreme thirst** separates dehydration from diabetes on whether hunger and weight loss accompany it. **Muscle weakness** separates the two thyroid conditions on the *direction* of change — weight gain with a slow heart rate against weight loss with a fast one.

Loss of Appetite

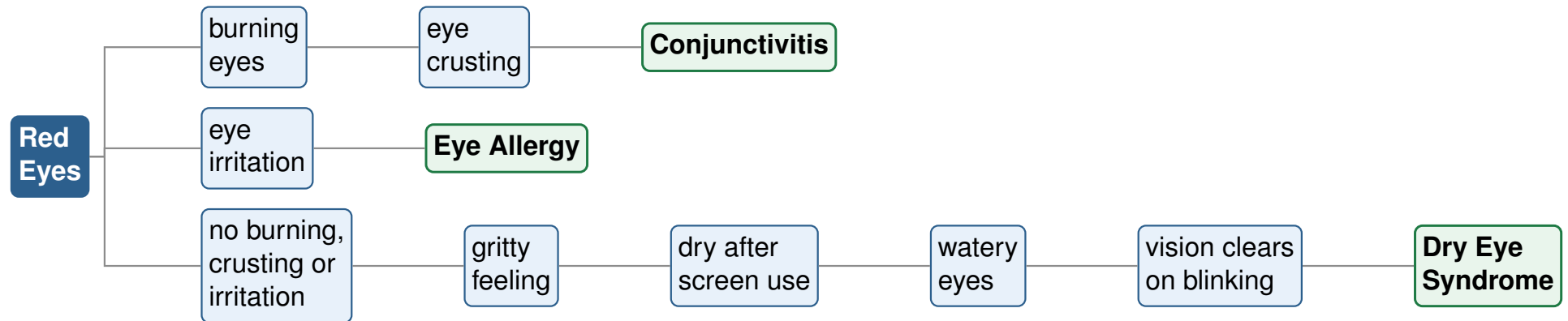
No fever, no short breath, no fatigue | 3 conditions



Gastritis and peptic ulcer are separated by symptom *severity* rather than symptom identity — nausea against mild nausea, vomiting against severe vomiting — which makes this the branch most sensitive to how consistently the interview captures intensity.

Red Eyes

Eye branch, evaluated independently of fever | 3 conditions



The only branch reached by explicit *exclusion*: dry eye syndrome is entered when burning, crusting, and irritation are all absent. Note the asymmetry — eye allergy resolves after a single symptom, while dry eye syndrome requires five. This branch is also the only one that may open alongside a fever branch in the same consultation.